

MINIMUM VIABLE HUMANOID · COMPONENT LIST AND COST ESTIMATE

JX0

walks · talks · 17 STS3215 joints



46.0 cm

HEIGHT

2.85 kg

MASS

17

STS3215 JOINTS

20

PRINTED PARTS

28

BOM LINES

PARTS FOR ONE ROBOT

₹58,234

THE 17 JOINT SERVOS

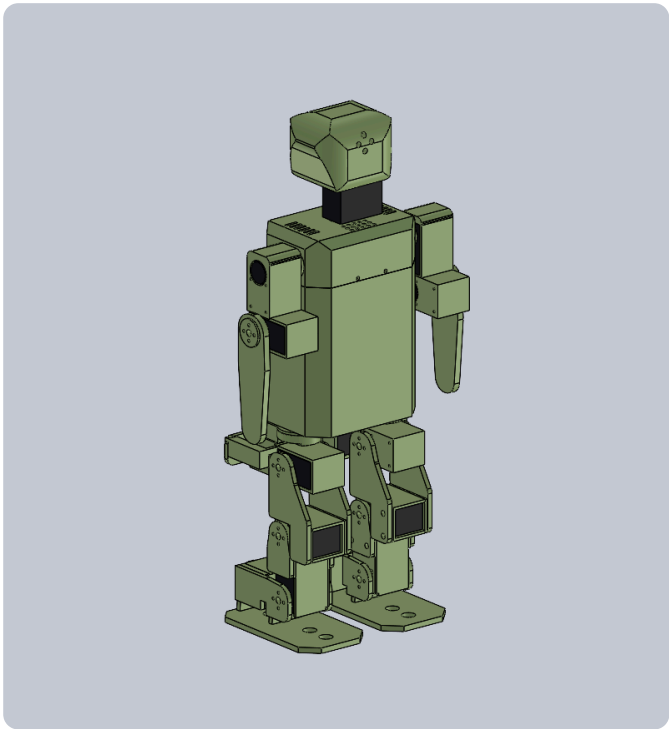
₹41,123

1. What JX0 is

JX0 is a small humanoid robot designed to be built by one student at home, from a 3D printer and parts that can be ordered online in India. Its look and its joints follow a friend's working robot: a faceted green body, every joint a 12 V STS3215. It is the working prototype of the JX1 project: it uses the same design pipeline (gait planner, servo sizing, physics simulation) at a price a student can afford.

Specification

Height / mass	46.0 cm / 2.85 kg (CAD, all parts)
Joints	17, every one a Feetech STS3215 12 V serial bus servo (30 kg·cm stall, position feedback), like the reference robot
Legs	6 joints each; every pitch and roll joint double-sided (a U-bracket on the servo's horn and rear hub, the body in a cage), like the reference robot
Arms and head	shoulder pitch and elbow per arm, a paddle blade; neck yaw under a soft rounded head
Brain	Raspberry Pi 4: 100 Hz gait playback with arm swing, IMU balance, offline speech recognition and voice
Talking	Claude over Wi-Fi writes the replies and calls the robot's actions (walk, turn, wave, nod, look)
Power	3S 2200 mAh LiPo (about 73 min of walking, CALCULATED) or a 12.6 V bench supply
Structure	20 parts, 823 g on a home 3D printer: pelvis and leg parts in PETG, printed near-solid; body, head and arms in pastel green matte PLA



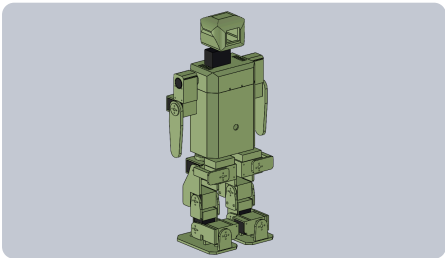
Done (on the computer)

- ✓ SolidWorks CAD: 20 printable parts and the full assembly with all 17 servos, built by script
 - ✓ Servo sizing all pass (tightest 1.47×); every printed leg part passes a finite-element check
 - ✓ Walking in a physics simulation of the CAD robot: 14 of 14 gaits pass, arms swinging
 - ✓ No part collisions in 1,373 moving poses; all 140 walks with random model errors and every push up to 0.96 N·s stay up
 - ✓ Robot program (voice, Claude, walking, gestures) runs on the simulated robot: 24/24 missions
- Next: buy the parts → print → assemble → first steps

How this estimate was made

- Every line is a real product page at an Indian store (Robu, Evelta, Robocraze, ThinkRobotics, Quartz Components and others), price checked 24 Sep – 5 Oct 2026.
- GST is included where the store states it; shipping is not included.
- Seven small lines (PETG, three screw lengths, PTFE tape, button, jumper wires) are ESTIMATED.
- Over the original ₹50,000 target because all 17 joints are STS3215, as on the reference robot.

From CAD to a working robot



SolidWorks assembly, back view



waving (simulation, the robot's own program)



walks into a bottle, as in the reference video

2. Cost summary

PARTS TOTAL

₹58,234

28 line items, one robot

JOINT SERVOS

71 %

of the cost: 17 × STS3215 at ₹2,419

EVERYTHING ELSE

₹17,111

brain, power, voice, structure

By group



Groups

GROUP	WHAT	₹	SHARE
Servos	17 × STS3215 (legs, arms, neck)	₹41,123	70.6 %
Electronics	Raspberry Pi 4, microSD, servo driver, wiring	₹9,219	15.8 %
Structure	filament, screws, inserts, foot rubber	₹4,074	7.0 %
Power	LiPo, charger, 5 V converter, switch, alarm	₹3,122	5.4 %
Voice	microphone, amplifier, speaker	₹569	1.0 %
Sensors	IMU	₹127	0.2 %
Total		₹58,234	100 %

Five biggest lines

ITEM	MODEL	QTY	₹
Joint servo (all 17 joints: 12 legs, 4 arms, neck)	Feetech STS3215 12 V 30 kg-cm serial bus servo (sold as ST3...	17	₹41,123
Brain	Raspberry Pi 4 Model B 2 GB (with case)	1	₹7,000
Battery	Bonka 3S 11.1 V 2200 mAh 35C LiPo	1	₹2,039
microSD card	SanDisk Ultra 32 GB A1	1	₹1,499
Filament for the body, head and arms (the reference robot's pastel green)	eSun PLA-Matte 1.75 mm 1 kg, Mint Green	1	₹1,299

Why the STS3215 everywhere: the walking needs up to 2.0 N·m at the hip roll (with a 1.25–1.5× safety margin), more than MG996R-class hobby servos give; the serial bus servos also report their position, which the calibration and the balance loop use; and one servo type on one bus is how the reference robot is built.

3. Complete list of items

#	ITEM	STORE	QTY	UNIT	TOTAL
01	Joint servo (all 17 joints: 12 legs, 4 arms, neck) Feetech STS3215 12 V 30 kg-cm serial bus servo (sold as ST3215-C018)	Evelta	17	₹2,419	₹41,123
03	Bus servo driver Waveshare Serial Bus Servo Driver board	ThinkRobotics	1	₹500	₹500
04	Brain Raspberry Pi 4 Model B 2 GB (with case)	ThinkRobotics	1	₹7,000	₹7,000
05	microSD card SanDisk Ultra 32 GB A1	Quartz Components	1	₹1,499	₹1,499
06	IMU MPU6050 gyro + accelerometer	Quartz Components	1	₹127	₹127
07	Microphone INMP441 I2S MEMS microphone	Quartz Components	1	₹145	₹145
08	Amplifier MAX98357A I2S 3 W class-D amplifier	Quartz Components	1	₹124	₹124
09	Speaker Waveshare 2030 cavity speaker 8 Ω 2 W	ThinkRobotics	1	₹300	₹300
12	Battery Bonka 3S 11.1 V 2200 mAh 35C LiPo	Robocraze	1	₹2,039	₹2,039
13	Charger B3 balance charger for 2S/3S LiPo	Quartz Components	1	₹294	₹294
14	5 V supply for the Pi UBEC 5 V 5 A (2-8S input)	Robu	1	₹409	₹409
16	Battery connector XT60 male/female pair	Robocraze	2	₹29	₹58
17	Power switch 10 A DC-rated toggle switch (CALONIX)	Quartz Components	1	₹63	₹63
18	Power wire 16 AWG silicone wire (per metre)	Robu	2	₹49	₹98
19	Signal wire 20 AWG silicone wire (per metre)	Robu	3	₹21	₹63
20	Battery alarm 1-8S LiPo voltage checker / low-voltage buzzer	Quartz Components	1	₹98	₹98
21	Filament for the body, head and arms (the reference robot's pastel green) eSun PLA-Matte 1.75 mm 1 kg, Mint Green	Robocraze	1	₹1,299	₹1,299
21b	Filament for the pelvis and leg brackets PETG 1.75 mm 1 kg, green (Quartz Components PETG Pro)	Quartz Components	1	₹759	₹759
22	M2 x 6 screws (servo cases, hip-roll horns) M2 x 6 12.9 socket head (pack of 10)	Indian Hobby Center	7	₹49	₹343
22b	M2 x 8 screws M2 x 8 12.9 socket head (pack of 10)	Indian Hobby Center	4	₹49	₹196
22c	M2 x 10 screws M2 x 10 12.9 socket head (pack of 10)	Indian Hobby Center	6	₹49	₹294
22d	M2 x 12 screws M2 x 12 12.9 socket head (pack of 10)	Indian Hobby Center	1	₹49	₹49
23	M3 x 8 screws (torso to pelvis) M3 x 8 socket head SS202 (pack of 10)	Quartz Components	1	₹35	₹35
24	M3 heat-set inserts M3 x 4 brass heat-set inserts (25 pcs)	Robu	1	₹91	₹91
25	Foot grip Natural rubber sheet 1 mm x 100 mm x 1 m	IndustryBuying	1	₹978	₹978
25b	Hip-yaw thrust rings PTFE thread-seal tape, 1 roll	any hardware shop	1	₹30	₹30
26	Push-to-talk button 12 mm momentary push button	any electronics shop	1	₹20	₹20
27	Jumper wires and servo extensions Dupont jumpers + 3-pin extensions	any electronics shop	1	₹200	₹200

Total for one JX0

₹58,234

Rupees Fifty-Eight Thousand Two Hundred Thirty-Four only